

PORT EN BESSIN, France

WMO: 070290

Lat: 49.350N

Lon: 0.767W

Elev: 73

StdP: 100.45

Time Zone: 1.00 (EUC)

Period: 86-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.9	(c) -1.0	(d) -7.0	(e) 2.1	(f) -1.9	(g) -4.7	(h) 2.6	(i) 0.3	(j) 17.4	(k) 6.8	(l) 15.8	(m) 6.8	(n) 4.0	(o) 120	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.1	(c) 24.1	(d) 19.0	(e) 22.1	(f) 18.4	(g) 20.8	(h) 17.6	(i) 20.0	(j) 22.7	(k) 19.0	(l) 21.3	(m) 18.2	(n) 20.2	(o) 4.6	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.1	(b) 14.0	(c) 21.4	(d) 18.1	(e) 13.1	(f) 20.1	(g) 17.4	(h) 12.5	(i) 19.4	(j) 57.6	(k) 22.9	(l) 54.2	(m) 21.5	(n) 51.7	(o) 20.3	(p) 23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.0	(b) 13.2	(c) 11.7	(e) -5.3	(f) 28.5	(g) 3.9	(h) 2.7	(i) -8.1	(j) 30.5	(k) -10.5	(l) 32.0	(m) -12.7	(n) 33.5	(o) -15.5	(p) 35.5		
(a) 15.0	(b) 13.2	(c) 11.7	(e) -6.1	(f) 21.5	(g) 3.6	(h) 1.3	(i) -8.8	(j) 22.4	(k) -10.9	(l) 23.2	(m) -13.0	(n) 23.9	(o) -15.6	(p) 24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	5.9	6.1	7.7	9.4	12.1	14.6	16.5	17.0	16.0	13.0	9.0	6.7		
	DBStd	4.87	3.36	3.66	3.07	2.59	2.28	2.43	1.87	2.12	2.42	2.58	2.62	3.20		
	HDD10.0	538	130	114	84	41	5	0	0	0	0	6	49	109		
	HDD18.3	2635	386	343	331	267	194	115	64	52	76	165	281	361		
	CDD10.0	974	2	5	13	23	70	139	201	217	181	99	18	7		
	CDD18.3	29	0	0	0	0	1	4	6	11	7	1	0	0		
	CDH23.3	102	0	0	0	0	9	17	12	41	21	2	0	0		
	CDH26.7	21	0	0	0	0	2	4	1	10	4	0	0	0		
(14) Wind	WSAvg	5.8	6.8	7.1	6.3	6.1	5.3	4.9	4.9	4.9	5.2	5.7	6.2	6.7		
(15) Precipitation	PrecAvg	847	85	69	57	60	57	56	55	62	62	95	98	97		
	PrecMax	1108	196	140	171	148	126	139	120	136	120	181	193	228		
	PrecMin	620	16	12	14	14	17	16	14	14	8	13	26	19		
	PrecStd	124	43	39	36	32	28	29	28	37	31	41	41	47		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	15.0	17.8	19.2	23.3	25.4	25.2	28.3	26.1	22.0	16.5	14.4		
		MCWB	11.7	11.6	12.7	13.9	16.4	19.5	20.2	19.5	19.4	16.8	14.6	13.0		
	2%	DB	12.3	13.3	14.9	16.5	19.1	21.8	22.6	24.0	22.3	19.2	14.6	13.2		
		MCWB	11.0	11.0	11.8	12.6	15.6	17.8	18.8	19.5	18.3	15.9	12.9	12.0		
	5%	DB	11.3	12.2	13.4	14.7	17.1	19.9	21.1	21.8	20.7	17.6	13.6	12.2		
		MCWB	10.0	10.4	11.2	11.7	14.3	16.8	18.1	18.6	17.7	15.2	12.1	11.0		
	10%	DB	10.3	11.0	12.1	13.2	15.7	18.3	19.9	20.4	19.5	16.5	12.8	11.2		
		MCWB	9.2	9.6	10.4	11.0	13.2	15.8	17.4	17.5	17.0	14.6	11.5	10.1		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.5	13.7	14.8	18.0	20.7	20.9	21.4	20.6	17.7	15.1	13.5		
		MCDB	12.8	14.1	17.0	17.6	21.4	24.1	23.5	24.8	23.0	20.3	15.8	14.1		
	2%	WB	11.2	11.6	12.3	13.2	15.8	18.4	19.5	20.1	19.1	16.4	13.2	12.2		
		MCDB	12.1	12.8	14.3	15.6	18.5	21.0	21.9	22.8	21.4	18.3	14.1	13.1		
	5%	WB	10.2	10.6	11.5	12.1	14.6	17.1	18.5	19.0	18.2	15.6	12.5	11.2		
		MCDB	11.1	11.9	13.1	14.2	16.6	19.3	20.6	21.2	20.0	17.2	13.4	12.1		
	10%	WB	9.3	9.7	10.5	11.2	13.6	16.2	17.6	18.2	17.5	14.9	11.6	10.2		
		MCDB	10.2	11.0	11.9	12.9	15.4	18.2	19.4	20.2	19.2	16.3	12.7	11.1		
(36) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.2	5.0	5.3	5.3	5.8	5.7	6.1	5.8	5.3	4.7	4.1		
		MCDBR	4.5	5.7	7.2	7.3	8.1	8.6	7.6	8.4	7.5	6.8	5.2	4.4		
	5% WB	MCWBR	4.1	4.2	4.9	4.6	5.1	5.1	4.4	4.8	4.5	4.3	4.3	4.0		
		MCDWR	4.3	4.6	6.7	6.5	7.6	7.6	6.8	7.2	6.7	6.2	4.8	4.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.329	0.346	0.379	0.395	0.401	0.398	0.392	0.386	0.373	0.359	0.335	0.323		
		taud	2.425	2.399	2.305	2.282	2.310	2.350	2.369	2.406	2.426	2.453	2.448	2.393		
	Ebn at Noon	Ebn at Noon	719	796	824	850	859	863	862	851	825	770	705	666		
		Edh at Noon	65	83	107	122	123	119	115	106	94	78	63	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.90	1.63	2.82	4.42	5.22	5.81	5.60	4.66	3.59	2.02	1.09	0.73		
	RadStd	RadStd	0.08	0.19	0.30	0.39	0.41	0.57	0.36	0.32	0.23	0.17	0.13	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page